

Enhancing Quality and Efficiency in Copper Clad Laminate Manufacturing: A Data-Driven Approach to Defect Reduction and Process Optimization

EGR 522: Stats Quality Control

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Abstract

Copper Clad Laminates (CCLs) are critical materials in the production of printed circuit boards (PCBs), serving as the conductive and structural foundation for electronic devices. The manufacturing process for CCLs is intricate, involving stages such as resin impregnation, lamination, and finishing, where minor variations can significantly impact quality and yield. This project addressed key challenges in CCL manufacturing, particularly the current rejection rate of 1.29%, which exceeds the industry target of 0.5%. Defects such as voids, bubbles, delamination, and High Potential (Hi-Pot) test failures were identified as primary contributors to scrap.

Using a combination of statistical tools and quality control methodologies, including Pareto analysis, Fishbone diagrams, Control Charts, and the seven QC tools, the project systematically analyzed data to uncover the root causes of defects and inefficiencies. Automated Optical Inspection (AOI), thermocouples, and destructive testing methods were employed to monitor critical parameters and identify process variability. Value-added and non-value-added time analysis highlighted inefficiencies in material handling and stage transitions, particularly in receiving and post-treating stages.

The findings informed actionable recommendations to optimize resin distribution, improve curing consistency, enhance defect detection systems, and streamline workflow transitions. The project demonstrated the effectiveness of a data-driven approach to defect reduction, achieving measurable progress toward meeting the target scrap rate. These results underscore the potential for continuous process improvement to enhance product reliability, reduce costs, and support long-term manufacturing excellence in the CCL industry.

Introduction

Background

Copper Clad Laminates (CCLs) are foundational materials used in the production of printed circuit boards (PCBs) and other electronic components. These laminates consist of a thin layer of copper bonded to a substrate, such as fiberglass-reinforced epoxy resin, which serves as the base material for circuit boards. CCLs provide the conductive surface and mechanical strength needed for PCB manufacturing, making them critical to the performance and reliability of electronic devices.

The manufacturing process for CCLs is complex, involving multiple stages where minor variations in parameters can significantly affect product quality and yield. The process begins with the impregnation of resin into a fiberglass substrate (prepreg manufacturing) and continues with lamination, where the resin-saturated fiberglass is bonded with copper under precise conditions of temperature, pressure, and time. The resulting CCLs must meet stringent quality standards to ensure uniform thickness, proper adhesion, and consistent electrical and mechanical properties.

Defects such as improper resin distribution, voids, bubbles, delamination, and thickness inconsistencies are common challenges in CCL manufacturing. These defects not only increase scrap rates but also compromise the reliability of downstream products, resulting in higher production costs and reduced customer satisfaction. Achieving consistent quality requires a detailed understanding of the process, rigorous quality control, and continuous improvement.

Problem Statement

The copper-clad laminate (CCL) manufacturing process is currently facing challenges with customer rejection rates of 1.29% of issued materials; exceeding customer performance targets of 0.5%, as captured by customer Automated Optical Inspection (AOI). The primary defects contributing to rejections/scrap during AOI are voids, foreign objects, and surface defects. Additional rejections are attributed to high potential test (Hi-Pot) failures. As of September 2024, current year-to-date (YTD) averages of customer rejections show that Hi-Pot failures account for 38.56%, surface defects for 35.37%, voids for 19.56%, and foreign materials defects for 6.51% significantly affecting the ability to meet customer standards.

Predicted root causes of these defects include environmental contamination, process inefficiencies, improper resin distribution, and insufficient process control. These issues result in high rejection/scrap rates, increased production costs, and compromised product quality. The current quality control methods have not fully resolved these issues, requiring further evaluation of the manufacturing process and using existing quality control tools.

Objectives

1. **Evaluate the Current Process:** Conduct a comprehensive analysis of the CCL manufacturing process, including prepreg production and copper lamination, from raw material inputs to the final product.
2. **Identify Process Steps:** Map the entire manufacturing workflow, highlighting critical stages where defects are most likely to occur.
3. **Analyze Quality Parameters:** Investigate critical metrics such as resin distribution, curing conditions (temperature and time), pressure uniformity, and thickness control.
4. **Implement Quality Control (QC) Tools:** Apply QC tools such as control charts, Pareto analysis, Fishbone diagrams, and Failure Mode and Effects Analysis (FMEA) to identify and address defects systematically.
5. **Improve Detection Methods:** Assess the capabilities of existing defect detection systems and propose enhancements to identify issues earlier in the process.
6. **Develop a Model-Based Design:** Integrate real-time, in-process quality checks and feedback mechanisms to ensure continuous process control and improvement.
7. **Provide Recommendations for Improvement:** Deliver actionable recommendations for process optimization, defect reduction, and improved detection accuracy.

Process Overview

The production of copper-clad laminates (CCLs) is a precise and intricate process that begins with the preparation of raw materials, including glass fabric, resin, and copper foil. The process starts with the compounding stage, where the resin system components are combined. In this stage, the epoxy resin is carefully blended with curing agents, hardeners, and other additives to achieve the desired mechanical, electrical, and thermal properties. The resin system must be mixed with precision to ensure consistency and homogeneity, as any variations can impact the performance and reliability of the final product. Once the resin system is prepared, it is ready for the next stage in the process.

Following compounding, the process moves to the impregnation stage, where the glass fabric is passed through a treater tower to be coated with the resin. The resin-saturated fabric, known as prepreg, is partially cured to a "B-stage," making it tack-free but still thermosettable for further processing. The prepreg is then prepared in the ply-up stage, where it is sheeted off from continuous rolls into large sheets. These sheets are organized and prepared according to specific product compositions, accommodating variations in resin formulation, reinforcement types, and thicknesses required for different CCL specifications.

Next, in the lay-up stage, alternating layers of prepreg sheets and copper foil are carefully stacked together to form a layered composite. Precise alignment during this stage ensures uniformity and consistency in the final laminate. The stacked materials are then subjected to the pressing stage, where high temperature and pressure are applied to fully cure the resin and bond the copper foil to the prepreg layers. This results in a solid composite material with uniform thickness and strong

adhesion between layers. After pressing, the laminated sheets are cut into smaller panels during the break-down stage and are inspected for defects such as voids, bubbles, or delamination.

In the finishing stage, the panels undergo a series of essential steps to prepare them for final use and ensure quality. First, the laminated sheets are resized into smaller, standardized panels through precise cutting, ensuring they conform to the required dimensions. This is followed by edge treatment, where the panel edges are smoothed and trimmed to remove any burrs or irregularities, improving their safety and compatibility with downstream processes. Each panel is then subjected to part labeling, where identification marks, such as lot numbers or specifications, are applied to facilitate traceability throughout the supply chain. Finally, the panels undergo rigorous final quality inspections to identify and reject any defects, such as surface imperfections, delamination, or thickness inconsistencies, ensuring that only products meeting strict quality standards are delivered to customers.

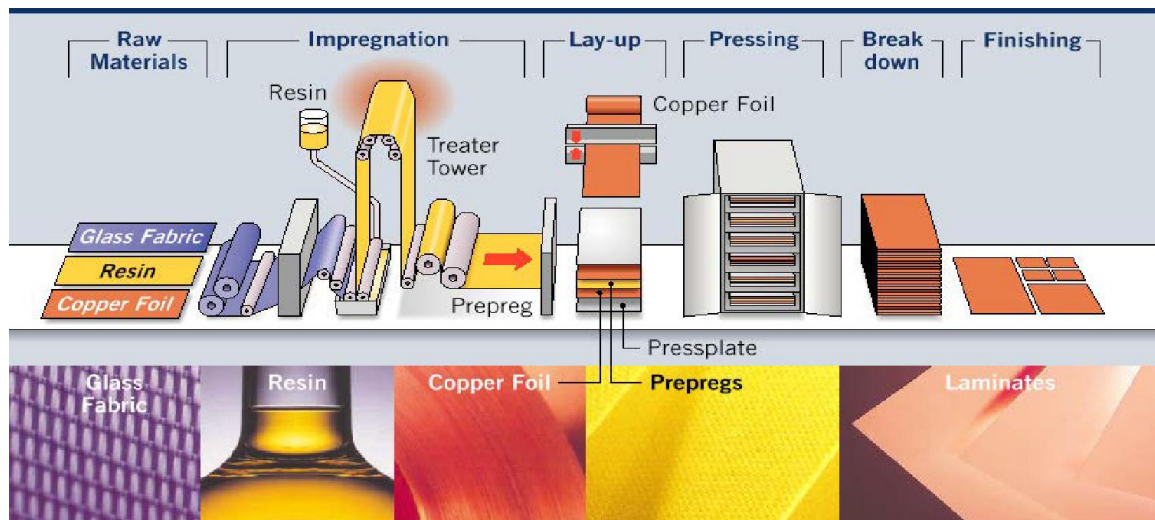


Figure 1: Process flow for CCL manufacturing

This comprehensive and methodical process, illustrated in Fig. 1., ensures that the final copper-clad laminates deliver the mechanical strength, electrical performance, and reliability required for high-performance applications in the electronics industry. From the precise compounding of resin systems to final finishing, each step is critical to achieving the high standards demanded by advanced electronic manufacturing.

Process Quality Control Metrics

The copper-clad laminate (CCL) manufacturing process involves rigorous quality control at every stage to ensure product reliability, consistency, and adherence to customer specifications. Below is an outline of the quality control (QC) metrics applied at each stage:

Compounding Stage

Objective: Ensure the resin system components are precisely mixed to achieve the desired properties.

Key QC Metrics:

- **Component Ratios:** Accurate measurement of resin, curing agents, and additives.
- **Homogeneity:** Verification of uniform mixing using viscosity testing.
- **Storage Stability:** Testing resin for pot life and gel time to ensure it remains usable during downstream processing.

Impregnation Stage

Objective: Coat glass fabric evenly with resin and partially cure it to a “B-stage” state.

Key QC Metrics:

- **Resin Pickup:** Measurement of the resin weight percentage absorbed by the glass fabric.
- **Coating Uniformity:** Inspection for consistent resin distribution across the fabric.
- **Curing Properties:** Monitoring partial curing levels to ensure the prepreg achieves the correct tack-free state.
- **Foreign Material & Cosmetic Defect Detection:** Monitoring via automated optical inspection for variance in resin uniformity or contrast related to the inclusion of foreign material.
- **Thickness Control:** Ensuring prepreg sheets meet thickness specifications.

Ply-Up Stage

Objective: Sheet prepreg rolls into large panels and prepares them for lay-up.

Key QC Metrics:

- **Dimensional Accuracy:** Verification of sheet dimensions to ensure proper fit during lay-up.
- **Sorting and Composition Verification:** Checking for correct material composition (e.g., resin type, reinforcement type, thickness).
- **Surface Integrity:** Inspection of prepreg sheets for surface defects, such as wrinkles or voids.

Lay-Up Stage

Objective: Stack layers of prepreg and copper foil into a composite assembly with precise alignment.

Key QC Metrics:

- **Layer Alignment:** Ensuring precise stacking to prevent misalignment during pressing.
- **Material Compatibility:** Verifying the prepreg and copper foil layers match product specifications.
- **Contaminant Inspection:** Checking for foreign particles that could cause defects in the final laminate.

Pressing Stage

Objective: Bond layers under high temperature, vacuum, and pressure to form a solid composite material.

Key QC Metrics:

- **Temperature and Pressure Profiles:** Continuous monitoring of press parameters to ensure optimal curing and bonding.
- **Bond Integrity:** Destructive testing for peel strength to evaluate adhesion between copper and prepreg.
- **Thickness Uniformity:** Measurement of laminate thickness to ensure it meets specifications.

Break-Down Stage

Objective: Cut the laminated sheets into smaller panels and inspect them for defects.

Key QC Metrics:

- **Dimensional Accuracy:** Verification of panel dimensions using automated measurement tools.
- **Defect Inspection:** Visual and automated checks for surface defects, delamination, or thickness variations.
- **Edge Quality:** Inspection for burrs or irregularities from cutting processes.

Finishing Stage

Objective: Prepare panels for final delivery, including sizing, edge treatment, labeling, and quality inspections.

Key QC Metrics:

- **Panel Dimensions:** Final verification of panel size and thickness.
- **Edge Treatment Quality:** Checking for smoothness and absence of burrs or rough edges.
- **Traceability:** Ensuring accurate and clear application of lot numbers or other labels.
- **Final Surface Inspection:** Detailed visual and automated inspections for cosmetic and functional defects.
- **Copper Type Validation:** Ensures proper characterization of products.
- **Adherence to Standards:** Cross-checking panels against industry standards (e.g., IPC standards for laminates).

Key Quality Attributes and Potential Defects

Each stage of copper-clad laminate (CCL) manufacturing requires careful control of key quality attributes to ensure reliable, high-performance products. Below is an outline of these quality attributes and potential defects, emphasizing their impact on product performance.

Resin System Quality

Key Quality Attributes:

- Uniform composition of resin, curing agents, and additives.
- Controlled viscosity for proper impregnation of the glass fabric.
- Accurate curing properties, such as gel time and pot life.
- High adhesion strength to ensure proper bonding to glass fabric and copper foil.

Potential Defects:

- **Inconsistent Resin Mix:** Leads to poor mechanical and dielectric properties.
- **High or Low Viscosity:** Causes uneven impregnation or coating of the glass fabric.
- **Premature Curing:** Results in resin hardening before impregnation.
- **Weak Adhesion:** Causes delamination or void formation.
- **Foreign Material in Resin:** Contaminants like dust or debris can weaken adhesion or create voids in the final product.

Prepreg Quality

Key Quality Attributes:

- Precise resin-to-glass ratio for uniform dielectric properties.
- Coating uniformity across the glass fabric.
- Thickness consistency between and across prepreg sheets.
- Correct partial curing ("B-stage") for tack-free handling and further processing.

Potential Defects:

- Resin Starvation or Excess: Leads to voids or excessive resin bleed-out during pressing.
- Coating Non-Uniformity: Causes weak spots or thickness variations.
- Improper Curing: Results in tackiness or insufficient bonding during pressing.
- Foreign Material Inclusion: Dust, fibers, or other particles embedded in the prepreg can cause uneven curing, voids, or surface defects.

Copper Foil Quality**Key Quality Attributes:**

- Smooth, defect-free surface for uniform lamination.
- Strong adhesion capability to bond with prepreg.
- Thickness uniformity for consistent electrical performance.
- High electrical conductivity for reliable signal transmission.

Potential Defects:

- Surface Scratches or Dents: Impacts electrical pathways and bonding quality.
- Oxidation: Reduces adhesion and conductivity.
- Thickness Variations: Causes uneven lamination or electrical performance issues.
- Foreign Material on Copper Surface: Dust, oil, or debris can prevent proper bonding and create defects like delamination.

Laminate Dimensional Accuracy**Key Quality Attributes:**

- Uniform thickness across the laminate.
- Panel flatness for ease of downstream processing.
- Dimensional stability to resist warping or deformation.

Potential Defects:

- Thickness Variations: Results in uneven layers and improper PCB functionality.
- Warping or Bowing: Causes handling difficulties and improper fit in PCB applications.
- Dimensional Inconsistencies: Leads to assembly and compatibility issues.
- Foreign Material Between Layers: Trapped particles during stacking can lead to uneven thickness and weak bonding.

Laminate Structural Integrity**Key Quality Attributes:**

- Void and bubble-free construction for strength and reliability.

- High adhesion strength between the copper foil and prepreg layers.
- Resistance to delamination under mechanical stress.

Potential Defects:

- Internal Voids or Bubbles: Weakens structural integrity and impacts reliability.
- Delamination: Causes layers to separate, resulting in failure during use.
- Weak Bonding: Reduces mechanical strength and durability.
- Foreign Material Trapped Inside Laminate: Dust or particles can cause voids, weak spots, and delamination.

Surface Quality

Key Quality Attributes:

- Smooth, defect-free surface for cosmetic and functional purposes.
- Clean copper surface free from pinholes, cracks, or oxidation.
- Burr-free and smooth edges after cutting.

Potential Defects:

- Surface Scratches or Dents: Impacts appearance and functionality.
- Pinholes or Cracks: Compromise electrical pathways and reliability.
- Rough or Burr-Edged Panels: Increase the risk of damage during handling or further processing.
- Foreign Material on Panel Surface: Dust or debris affects adhesion and can result in electrical shorts or defects in downstream PCB manufacturing.

Electrical Properties

Key Quality Attributes:

- Uniform dielectric constant (D_k) for predictable signal performance.
- Low loss tangent (D_f) to minimize energy dissipation.
- High insulation resistance to prevent leakage.
- Reliable conductivity of the copper layer.
- High Potential Testing (Hi-Pot): Verifies the integrity of the insulation by applying a high voltage to detect leakage currents or breakdown points.

Potential Defects:

- Dielectric Variations: Causes signal distortions or inconsistencies.
- High Loss Tangent: Reduces electrical efficiency.
- Poor Conductivity: Results in overheating or signal failure.

- Foreign Material Inclusion: Non-conductive particles in the laminate can disrupt electrical pathways or insulation properties.
- Hi-Pot Failures: Indicate insulation weaknesses, pinholes, or breakdown under high-voltage conditions.

Thermal and Mechanical Properties

Key Quality Attributes:

- High glass transition temperature (Tg) for thermal stability.
- Efficient thermal conductivity for heat dissipation.
- Resistance to flexural stress and mechanical deformation.
- Low coefficient of thermal expansion (CTE) to avoid warping or cracking.

Potential Defects:

- Low Tg: Causes deformation or delamination under high temperatures.
- Weak Flexural Strength: Leads to cracking or mechanical failure.
- Excessive Thermal Expansion: Results in warping during PCB assembly.
- Foreign Material Inclusion: Can create thermal hotspots or weak points in the laminate.

Traceability and Consistency

Key Quality Attributes:

- Clear lot identification for traceability.
- Repeatability in production for consistent quality.
- Compliance with industry standards (e.g., IPC-4101).

Potential Defects:

- Missing Traceability Information: Complicates defect tracking and quality control.
- Inconsistent Quality: Results in product variability and customer dissatisfaction.
- Foreign Material in Records: Mislabeling or contamination of batches leads to confusion in tracking.

Final Inspection & Compliance

Key Quality Attributes:

- Panels free from visible or hidden defects.
- Conformance to customer specifications for size, thickness, and electrical properties.
- Proper handling and packaging to prevent damage during transportation.

Potential Defects:

- Undetected Defects: Result in customer complaints and returns.

- **Improper Dimensions:** Lead to assembly issues in PCB manufacturing.
- **Poor Packaging:** Causes damage to panels during shipping.
- **Foreign Material in Packaging:** Dust or debris during packing can affect panel surfaces or edges.

By systematically monitoring these key quality attributes, manufacturers can ensure that defects are minimized, and product performance remains consistent and reliable. Each attribute is critical for ensuring the final product meets the high standards demanded by the electronics industry.

Data Collection & Analysis

Data Collection

Data collection for this analysis utilized a comprehensive approach, combining real-time monitoring, historical records, and direct observations to ensure accurate identification of inefficiencies and defects in the copper-clad laminate (CCL) manufacturing process. Advanced tools and methodologies were employed at critical production stages to capture key performance metrics and insights.

Methodology

Data Collection Methods

1. **In-Line Sensors:** Monitored real-time process parameters such as resin distribution, curing temperature, and pressure during critical stages like impregnation and pressing.
2. **Rheometry & Physical Characterization Equipment:** Used throughout the process to characterize material properties.
3. **Automated Optical Inspection (AOI):** Deployed during the impregnation stage to detect foreign material, resin flow inconsistencies, and surface defects. AOI provided real-time data to support immediate corrective actions.
4. **Thermocouples:** Used in the pressing stage to precisely track temperature profiles and ensure curing consistency, addressing risks of delamination and incomplete bonding.
5. **Manual Inspections:** Conducted at various stages to validate automated system outputs and identify defects like voids, delamination, and surface irregularities.
6. **Historical Process Data:** Included production records and defect rates to establish trends in process efficiency and quality performance over time.
7. **Destructive Testing:**
 - **Destructive Physical Analysis (DPA):** Evaluated the structural integrity and bonding quality of laminated sheets after pressing.

- **Destructive Material Testing (DMT):** Examined pressed CCL batches for defects like voids and bubbles, with automatic rejection for non-compliant samples.

Key Metrics Captured

1. **Defect Rates:** Identified and categorized defects at each stage, including bubbles, voids, delamination, and surface inconsistencies.
2. **Curing Profiles:** Monitored temperature and resin curing times to ensure optimal conditions in pressing and impregnation.
3. **Cycle Times:** Measured process durations for value-added and non-value-added activities to highlight inefficiencies.
4. **Material Quality:** Assessed resin and prepreg properties, including thickness uniformity and resin-to-glass ratios.

Applications of Collected Data

1. **Process Monitoring:** Enabled identification of real-time deviations in critical parameters, allowing for immediate corrective actions to minimize defects.
2. **Quality Assurance:** Supported validation of product consistency and compliance with industry standards.
3. **Root Cause Analysis:** Data from AOI, thermocouples, and destructive testing informed investigations into recurring defects and inefficiencies.
4. **Process Optimization:** Historical and real-time data-informed decisions to streamline transitions, improve scheduling, and enhance material handling.

Value-Added Time Analysis

The process flow diagram (Fig. 2.) outlines the production stages and their respective time intervals, emphasizing the contrast between value-added and non-value-added times in the copper-clad laminate (CCL) manufacturing process. This analysis reveals significant inefficiencies in stage transitions, offering clear opportunities for process optimization.

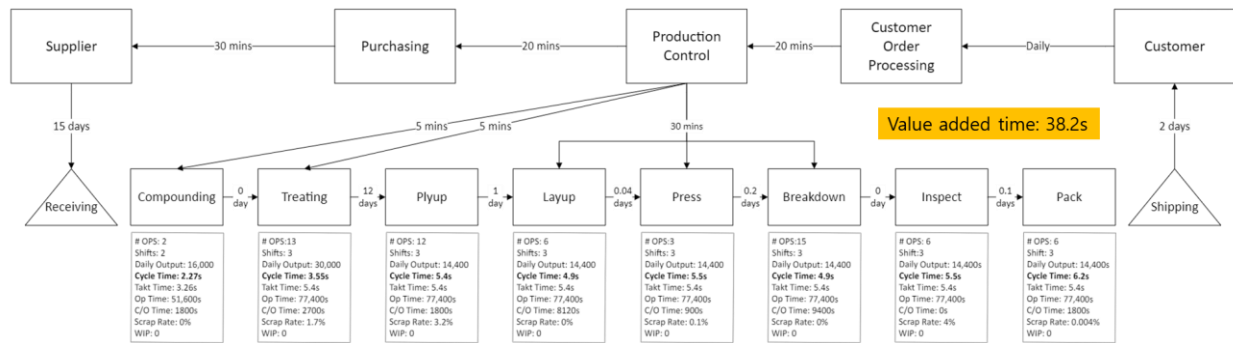


Figure 2: Value-Added Time Analysis

Findings

1. Cycle Times for Value-Added Processes:

- Compounding: 2.27 seconds
- Treating: 3.55 seconds
- Ply-Up: 5.4 seconds
- Lay-Up: 4.9 seconds
- Pressing: 5.5 seconds
- Break-Down: 4.9 seconds
- Inspection: 5.5 seconds
- Packaging: 6.2 seconds

2. Total Value-Added Time: 38.2 seconds

3. Non-Value-Added Delays:

- Receiving Stage: A 15-day wait for materials significantly extends the timeline.
- Post-Treating Stage: A 12-day delay before materials move to the next process further highlights inefficiencies.
- Transitions Between Stages: Cumulative delays in material movement and scheduling add substantial non-value-added time to the overall process.

Insights

The value-added time of 38.2 seconds is negligible compared to the overall timeline, which spans multiple days due to waiting and transition delays. Significant bottlenecks occur at the receiving and post-treating stages, where materials experience prolonged inactivity. While individual stages demonstrate high operational efficiency, the overall production flow is hindered by delays between processes.

Recommendations for Process Optimization

1. Streamline Material Transitions:

- Automate material handling and scheduling systems to reduce transition times between stages.
- Implement a just-in-time (JIT) inventory approach to minimize delays at the receiving stage.

2. Optimize Scheduling:

- Reduce idle time by synchronizing upstream and downstream processes.
- Introduce predictive scheduling tools to anticipate and address bottlenecks.

3. Monitor and Reduce Non-Value-Added Time:

- Analyze delays at each transition point and implement targeted solutions to streamline workflows.
- Use real-time tracking systems to ensure timely movement of materials.

High Potential (HiPot) Testing and Defect Analysis

The High Potential (HiPot) test is a critical quality control measure in CCL and PCB manufacturing, designed to verify the dielectric material's ability to withstand voltages higher than its rated voltage without breakdown. This stress test measures the dielectric strength of the PCB substrate and evaluates the insulation capability of the Device Under Test (DUT), ensuring reliability in high-performance applications.

HiPot Testing and Defect Contribution

1. **HiPot Defects:** As illustrated in **Figure 3**, HiPot-related defects are the leading cause of laminate scrap, highlighting the importance of effective insulation material and manufacturing consistency.

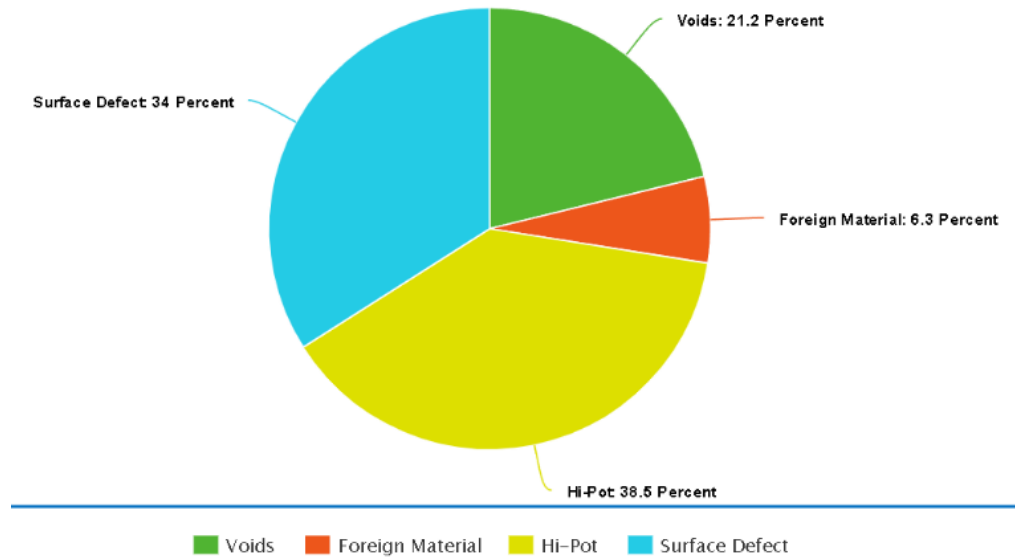


Figure 3: Pareto Defect Chart

2. **Surface Defects:** Surface defects, such as delamination, bubbles, or voids, follow closely behind HiPot as significant contributors to defect rates. These issues often stem from poor resin distribution, contamination, or inadequate curing during earlier stages.

Scrap Rate Performance

By week 23, a consistent downward trend in defect rates emerged, indicating that corrective actions and process optimizations were effectively addressing quality issues.

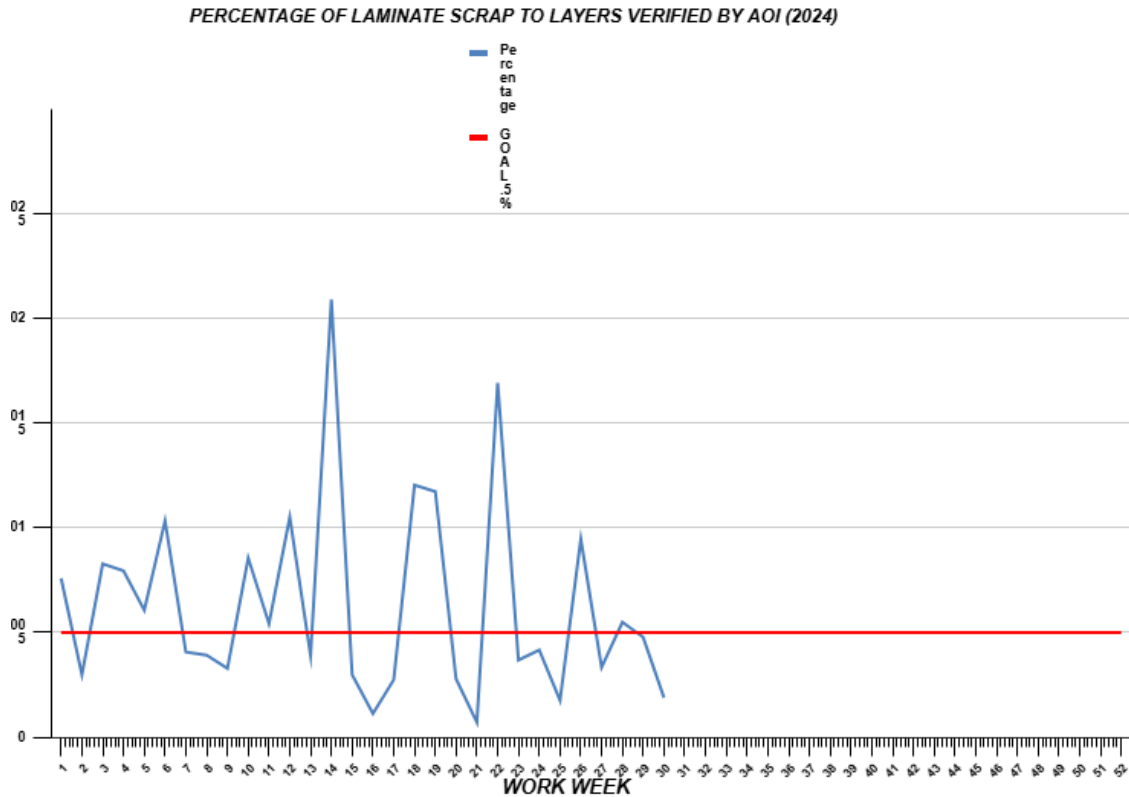


Figure 4: Laminate Scrap (0.5% Goal)

Observations & Insights

1. **Positive Trends:** The data demonstrates that the company's strategies for defect prevention—such as improving resin curing profiles, enhancing cleanliness, and refining HiPot testing protocols—are yielding significant results.
2. **Room for Improvement:** Despite overall progress, the occasional exceedance of the scrap rate target suggests opportunities for further process refinement, particularly in stages leading to HiPot and surface defects.

Recommendations

1. **Focus on HiPot-Related Defects:** Conduct a deeper analysis of material properties and insulation failures in laminates that do not pass the HiPot test. Enhance process controls in treating and pressing stages, where dielectric material quality is most influenced.
2. **Surface Defect Mitigation:** Strengthen contamination control measures during impregnation and lay-up to prevent voids and surface irregularities. Ensure curing profiles are consistently maintained to eliminate uneven resin flow or adhesion failures.
3. **Continued Monitoring:** Maintain detailed weekly tracking of defect trends to identify patterns and deviations early. Investigate anomalies during weeks with elevated scrap rates to refine preventive measures.

The HiPot test plays a pivotal role in ensuring the dielectric and insulation reliability of CCLs. The observed improvements in defect rates and adherence to the 0.5% scrap rate goal after week 23 reflect the success of implemented quality enhancements. By addressing remaining challenges, particularly in HiPot and surface defect areas, the company can further solidify its trajectory toward consistent production excellence.

Statistical and Quality Control Methods

To address the challenges in copper-clad laminate (CCL) manufacturing, particularly in reducing defects and optimizing processes, a combination of statistical and quality control tools was employed. These methods provided a structured approach to identifying root causes of defects, monitoring process variability, and implementing targeted improvements.

Statistical Analysis

Statistical tools were crucial in analyzing production data to uncover defect trends and their root causes:

1. **Pareto Charts** were used to prioritize defect types, revealing that voids, bubbles, and delamination accounted for the majority of scrap. This insight focused corrective actions on the most critical issues in stages such as resin treating and pressing.
2. **Fishbone Diagrams** (Fig. 5.) categorized potential causes into Material (e.g., resin inconsistencies), Method (e.g., curing variability), Machine (e.g., press calibration), and Environment (e.g., airborne debris). This systematic categorization helped address complex, multifactorial problems affecting resin flow and adhesion strength.

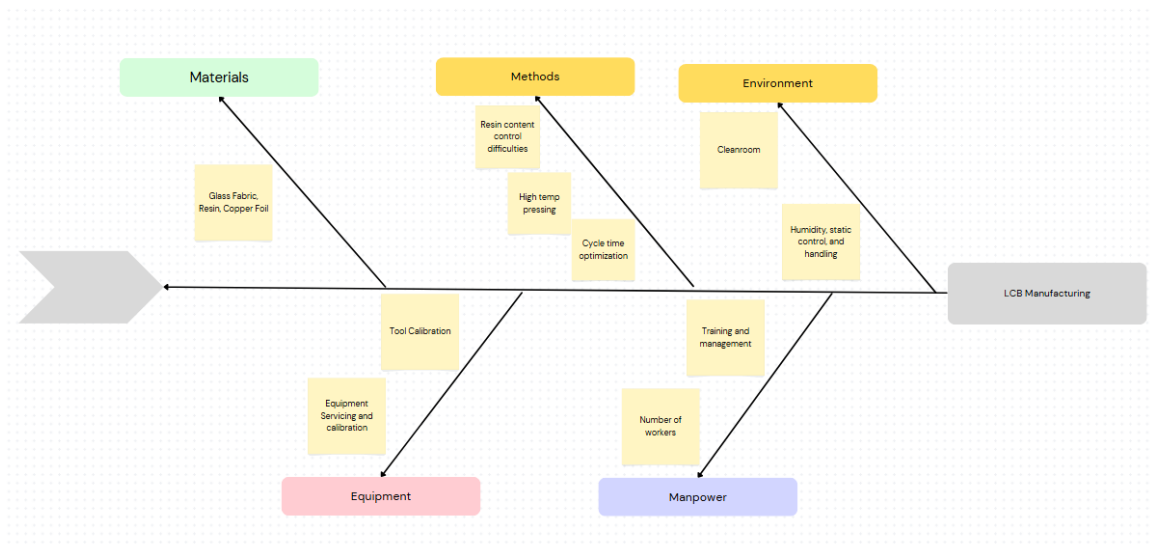


Figure 5: Fishbone Diagram

3. **Control Charts** monitored process stability, highlighting variability in curing temperature and pressure that correlated with increased delamination rates. By distinguishing between normal variation and out-of-control conditions, these charts supported proactive defect prevention.
4. **Regression Analysis** quantified relationships between process variables, such as curing time, pressure, and resin flow, providing actionable insights to optimize curing profiles and enhance laminate quality.

Application of 7 QC Tools

The 7 QC Tools were integrated into the project to systematically address quality challenges:

- **Pareto Charts** and **Fishbone Diagrams** identified and mapped the most frequent and impactful defect causes, guiding focused improvement efforts in critical stages like impregnation and pressing.
- **Histograms** visualized trends in resin distribution, laminate thickness, and pressing parameters, aiding in process consistency.
- **Control Charts** provided continuous monitoring of production stability, enabling early detection and correction of deviations.
- **Scatter Diagrams** revealed correlations between key variables, such as resin viscosity and laminate strength, informing adjustments to optimize product performance.
- **Check Sheets** tracked defect types and frequencies, uncovering patterns like recurring bubbles during resin treatment.
- **Flowcharts** mapped the end-to-end production workflow, identifying inefficiencies and bottlenecks, such as delays after treating and during material lay-up, and guiding streamlining efforts.

Impact on CCL Manufacturing

The combined use of statistical and QC tools directly addressed the primary challenges in CCL manufacturing. By focusing on critical defects, refining key parameters, and streamlining workflows, these methods enabled a data-driven approach to improving process reliability and reducing scrap. As a result, the tools provided actionable insights that enhanced production efficiency and supported long-term quality improvement.

Conclusion

The project addressed key challenges in copper-clad laminate (CCL) manufacturing by leveraging a structured approach to defect identification, process optimization, and quality improvement. Through detailed analysis and the application of advanced statistical and quality control tools,

significant insights were gained into the root causes of defects such as voids, bubbles, delamination, and Hi-Pot failures.

The findings demonstrated that strategic enhancements in process control, including better resin distribution, improved curing consistency, and contamination mitigation, could significantly reduce defect rates. The use of automated optical inspection (AOI), thermocouples, and destructive testing provided real-time and post-process insights that informed corrective actions.

Key outcomes included the identification of value-added and non-value-added time in the production flow, revealing inefficiencies in stage transitions, particularly in the receiving and post-treating stages. Recommendations for streamlined material handling, just-in-time inventory systems, and real-time monitoring systems were proposed to minimize delays and optimize throughput.

The project validated the importance of a data-driven approach to quality management, combining statistical tools such as Pareto charts, Fishbone diagrams, and Control Charts with the seven QC tools to reduce defects and enhance product reliability systematically.

Moving forward, continued monitoring, refinement of defect detection systems, and further process optimization will be critical in achieving consistent adherence to the 0.5% scrap rate goal and maintaining high standards in CCL manufacturing. These efforts will not only improve product quality and customer satisfaction but also drive operational efficiency and cost savings.

Team Roles and Contribution

Member	Role	Responsibility	Contribution
Timothy Liu	Writer/Editor	Research the basic background of the industry and create a more understandable narrative for the paper	Wrote the Abstract and Introduction. Helped outline the paper and created the basic sections of the paper.
Zerek Leyva	Data Analyst	Find the statistical problems and errors in the data set. Research the data collection methods and work closely alongside Josh.	Wrote the Data Collection and Analysis section of the paper. Created the slide presentation and helped condensed the information of the paper to transfer to the presentation/
Jeet Ramesh Khut	Data Analyst	Research the Quality Control Tools needed for the paper. Sourced the proper tools and software for the statistical graphs and data simplification overview.	Wrote the Statistical and Quality Control Methods. Created the charts and images for the paper. Researched the quality control tools needed for the paper.
Joshua Downing	Subject Matter Expert	Provide data from the customer and give insight into the customer operations and data collection. Provide the overall direction of the research paper.	Wrote the Conclusion of the paper. Gave insight to teammates and gave an understanding of the whole process on the customer's side. Created a link between the customer and the research team.
Suk Ki Lee	Writer/Editor	Create the outline of the paper and fill in the gaps that are needed to complete the paper. Provided an overview of what the paper needed and the flow of the sections.	Wrote the Process Overview section and the research paper along with the presentation slide. Edited the whole paper for proper alignment and errors.